

SignMuon, MuonSign, and the Role of Error Feedback

Anonymous submission

Abstract

SignMuon, the sign compression of the Muon update to one bit per parameter, is the most straightforward adaptation of Muon to an extremely low communication budget. It outperforms SignSGD in practice, yet it can diverge on a linear function. The natural conjecture attributes this to the order of operations, and holds that signing before rather than after the Linear Minimization Oracle (LMO) would remedy it. It does not: sign-before (MuonUSign) likewise diverges, as does signing on both sides (MuonSign), so that no naive placement of the sign around the LMO converges. Error feedback repairs one of them: EF21-MuonUSign attains the standard $\mathcal{O}(T^{-1/2})$ rate for smooth nonconvex optimization, and compressing the downlink as well exacts little beyond that. What the guarantee does exact is the placement. Across centralized, federated and language-modelling experiments the leading method is consistently sign-*after*-the-LMO, precisely the placement our counterexamples defeat and error feedback fails to repair; the provably convergent variants trail it. We document this tension rather than resolve it: on random instances the gradient/step alignment our counterexamples annihilate remains positive, so they constrain worst-case problems and not CIFAR-10.

1 Introduction

Training a deep network across clients consumes bandwidth as well as computation: every round each client transmits a full update, and under a limited link those updates dominate the wall-clock cost. Compression is the standard remedy (Bernstein et al. 2018, 2019; Beznosikov et al. 2023), and SignSGD is its extreme point: one bit per coordinate, at slight cost in accuracy. SignSGD, however, flattens each weight matrix into a vector, discarding the structure matrix-aware optimizers exploit. Muon exploits precisely that structure, orthonormalizing the momentum matrix before stepping, which stabilizes training and in several settings surpasses tuned adaptive methods (Jordan et al. 2024b; Bernstein and Newhouse 2024; Shah et al. 2025). What it transmits is a *dense* matrix at full precision, the one commodity a bandwidth-limited system cannot afford. Matrix geometry and a one-bit budget therefore pull in opposite directions.

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We study the natural ways to combine sign compression with the Muon LMO at one bit per parameter: **SignMuon**, which signs *after* the LMO, $\text{sign}(\text{LMO}(\cdot))$; **MuonUSign**, which signs *before*, $\text{LMO}(\text{sign}(\cdot))$; and **MuonSign**, which signs on *both* sides and so compresses uplink and downlink alike. All three retain Muon’s matrix-aware geometry, but only sign-*after* attains Muon’s accuracy: in federation the three span several accuracy points, in the order after, before, both sides, the last falling below SignSGD.

None of them, however, has a global convergence guarantee. We prove that all three can diverge: SignMuon on a 4×4 instance (Theorem 1), MuonUSign and MuonSign on a single common 5×5 one (Theorems 2–3); in each case the 1-bit step becomes an ascent direction on a linear objective. To restore convergence at the same communication volume we introduce **EF21-MuonUSign**, adapted from EF21-Muon (Gruntkowska et al. 2025), which error-feedbacks the *gradient* rather than the oracle’s output, and **EF21-MuonSign**, which adds a second loop on the downlink so that both channels are one bit. Both descend on the counterexamples above.

We evaluate on centralized CIFAR-10 with a ResNet-18, federated CIFAR-10 at $N = 11$ clients, the nanoGPT speedrun, and a synthetic convex quadratic. Our contributions are: (i) the three placements, as matrix-aware 1-bit optimizers; (ii) counterexamples proving all three may diverge (Theorems 1–3), at minimal matrix sizes; (iii) a universal divergence theorem (Theorem 4): error feedback on the LMO *output* repairs SignMuon for no (L, η, μ) ; (iv) EF21-MuonUSign and EF21-MuonSign, restoring the guarantee at a one-bit uplink and at one bit in both directions respectively; and (v) a tuned five-seed federated comparison of all six, which separates the three placements that single-seed evidence had shown as coincident and locates the guarantee’s cost at the placement, not the downlink. Proofs, algorithms and every numbered result cited but not stated here are in the supplementary material, which continues this numbering; each citation names its appendix section.

2 Related Work

Sign compression. SignSGD (Bernstein et al. 2018), extended to federated learning by majority vote (Bernstein et al. 2019), needs growing batches to converge; its bias is otherwise repaired by error feedback (Karimireddy et al. 2019),

sharpened into EF21 (Richtárik, Sokolov, and Fatkhullin 2021) and its bidirectional form EF21-P (Gruntkowska, Tyurin, and Richtárik 2023), by momentum (Cutkosky and Mehta 2020; Sun et al. 2023), or by randomizing the sign operator (Chen et al. 2020; Safaryan and Richtarik 2021; Jin et al. 2025). Coarser quantizers with unbiasedness or variance reduction are the other branch (Alistarh et al. 2017; Horváth et al. 2023; Beznosikov et al. 2023).

LMO optimizers and their limits. Muon (Jordan et al. 2024b) is the spectral-norm instance of a norm-constrained LMO step (Bernstein and Newhouse 2024; Pethick et al. 2025), analysed by Li and Hong (2025) and Kovalev (2025) and generalized to arbitrary layer norms by Gluon (Riabini et al. 2025); Crawshaw et al. (2025) survey the resulting family. Negative results exist for the uncompressed method too: Parshakova et al. (2026) show Muon need not converge on convex Lipschitz problems, and Shulgin et al. (2026) quantify the cost of an inexact oracle. Ours are of the same kind, but caused by the compressor rather than by the geometry.

Compressed and federated Muon. Gruntkowska et al. (2025) give the error-feedback framework for bidirectionally compressed Muon/Gluon that our positive result reduces to; Qian et al. (2026) add variance reduction, Takezawa et al. (2026) and Zhang and Gao (2025) study federated LMO steps without compression, Ahn et al. (2025) and Gupta et al. (2025) compress the oracle and its state respectively, and Thérien et al. (2025) report that Muon-produced updates compress well in practice. Concurrently, Mishra, Trivedi, and Kumar (2026) proposed plain SignMuon with an $\mathcal{O}(1/\sqrt{T})$ guarantee; Appendix A.6 establishes that their finalized rate covers only the gradient-sign instantiation, leaving the polar-sign method uninformative on precisely the instances of Theorem 1. Two further methods share our names but not our object: Bolatov et al. (2026) *alternate* spectral and sign steps rather than composing them, and Kravatskiy et al. (2025) mix the geometries *inside* the oracle, so that S-Muon remains an LMO for an explicit norm and retains the attendant theory. Our placements situate the sign *around* the oracle, where it constitutes a compressor and not a geometry, so the counterexamples below bear upon neither (Appendix A.7).

3 Problem Statement

We consider the stochastic optimization problem

$$\min_{\mathbf{X} \in \mathcal{X}} \{f(\mathbf{X}) := \mathbb{E}_{\xi \sim \mathcal{D}}[f_\xi(\mathbf{X})]\}, \quad (1)$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m \times n}$) and the $f_\xi : \mathcal{X} \rightarrow \mathbb{R}$ are continuously differentiable and possibly non-convex: $f_\xi(\mathbf{X})$ is the loss of a model $\mathbf{X}$ at a data point $\xi \sim \mathcal{D}$. In the federated setting the data are split across clients, and the training problem becomes

$$\min_{\mathbf{X} \in \mathcal{X}} \left\{ f(\mathbf{X}) := \frac{1}{N} \sum_{j=1}^N f_j(\mathbf{X}) \right\}, \quad f_j(\mathbf{X}) = \mathbb{E}_{\xi_j \sim \mathcal{D}_j} [f_j(\mathbf{X}; \xi_j)] \quad (2)$$

where $N \geq 1$ is the number of clients and f_j is the loss on the data $\mathcal{D}_j$ held by client $j \in [N] := \{1, \dots, N\}$.

Two assumptions run through the paper. The model is a tuple of matrix layers $\mathbf{X} = [\mathbf{X}_1, \dots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, with ∇_i the gradient block of layer i . Each layer carries the spectral norm $\|\cdot\|_{2 \rightarrow 2}$, whose dual is the nuclear norm $\|\cdot\|_*$, and we write $\|\nabla f(\mathbf{X})\|_*^2 := \sum_i \|\nabla_i f(\mathbf{X})\|_*^2$; a single matrix parameter is the case $p = 1$.

Assumption 1 (Lower boundedness) $f(\mathbf{X}) \geq f^*$ for all $\mathbf{X}$; where explicitly invoked, each local $f_j \geq f_j^*$ as well.

Assumption 2 (Layer-wise smoothness) For every layer i and all $\mathbf{X}, \mathbf{Y}$,

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \leq L_i^g \|\mathbf{X}_i - \mathbf{Y}_i\|_{2 \rightarrow 2} \quad (3)$$

for $g = f$ (constant L_i) and $g = f_j$ (constant $L_{i,j}$). A weaker layer-wise (L^0, L^1) form, stated in Appendix A.1, is used only in Corollary 2 (Appendix A.9).

A Linear Minimization Oracle (LMO) minimizes a linear form over the unit ball of a norm, returning the steepest-descent direction in that norm (Appendix A.1). Muon takes that norm to be spectral on each matrix layer, which turns the oracle into an orthogonalization: writing $\mathbf{M} = \mathbf{U}\Sigma\mathbf{V}^\top$ for the rank- r singular value decomposition of $\mathbf{M} \in \mathbb{R}^{m \times n}$, the LMO direction is

$$A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top \in \mathbb{R}^{m \times n}, \quad \text{polar}(\mathbf{M}) := -A(\mathbf{M}). \quad (4)$$

In practice the oracle is applied to a momentum estimate rather than to $\mathbf{G}_t$ itself; (5) below gives the full iteration. What Muon must transmit is the dense $\mathbf{D}_t$, precisely what a bandwidth-limited link cannot accommodate; our object is therefore its matrix-aware geometry at a one-bit budget.

4 Theory

From SignSGD to Sign A. Every method in the SignSGD lineage signs a *gradient*, but nothing in the construction requires it: any optimizer emitting a direction can have that direction signed instead, and should inherit whatever the direction was contributing. For LMO-based optimizers the expectation is more than an analogy, because their theory is uniform in the geometry: one analysis covers every norm, which enters only through its oracle and a pair of norm-equivalence constants (Pethick et al. 2025; Kovalev 2025; Riabini et al. 2025). If signing a direction were harmless it should be harmless across that family, and the member one actually seeks to compress is Muon.

A second reason to expect it to work is the one that misleads. The sign step is not merely compatible with the LMO view; it *is* an LMO, steepest descent in ℓ_∞ (Bernstein and Newhouse 2024), so signing a Muon direction composes two oracles, each sound alone. The composition is an LMO for *no* norm. An oracle for a norm $\|\cdot\|$ returns the direction $\mathbf{D} = -A(\mathbf{G})$, for which $\langle \mathbf{G}, \mathbf{D} \rangle = \max_{\|\mathbf{Y}\| \leq 1} \langle \mathbf{G}, \mathbf{Y} \rangle = \|\mathbf{G}\|_{\text{dual}}$, nonnegative because $\mathbf{0}$ is feasible; Theorems 1–3 exhibit gradients driving that inner product strictly negative. The guarantee is forfeited in the composition, not in either factor.

The Sign A family. Let A be any optimizer built on LMO directions. A Sign A method initializes $\mathbf{M}_0 = \mathbf{0}$ and at each iteration $t \geq 1$ performs

$$\begin{aligned} \mathbf{M}_t &= \mu \mathbf{M}_{t-1} + (1 - \mu) \mathbf{G}_t, \\ \tilde{\mathbf{M}}_t &= \mathbf{M}_t \text{ or } (1 - \mu) \mathbf{G}_t + \mu \mathbf{M}_t \text{ (Nesterov),} \\ \mathbf{D}_t &= -A(\tilde{\mathbf{M}}_t), \quad \mathbf{s}_t = \text{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m \times n}, \\ \mathbf{X}_t &= \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t, \end{aligned} \quad (5)$$

where $\mathbf{X}_t \in \mathbb{R}^{m \times n}$ is the parameter matrix, $\mathbf{G}_t$ a stochastic gradient evaluated at $\mathbf{X}_{t-1}$, $\mu \in [0, 1)$ the momentum coefficient and η_t the learning rate; $A(\cdot)$ is the LMO, and $\text{sign}(\cdot)$ acts element-wise on the structured direction $\mathbf{D}_t$, with $\text{sign}(0)$ resolved to an independent random ± 1 so that the transmitted alphabet stays binary. Momentum is in exponential-moving-average form throughout; the heavy-ball form differs by the positive factor $1/(1 - \mu)$, which $\text{sign}(\cdot)$ and the LMO both absorb, so no result below distinguishes them.

Our three methods are the three placements of the sign around this oracle. They share the update $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t$ and differ only in where the sign acts on $\tilde{\mathbf{M}}_t$:

$$\mathbf{s}_t = \begin{cases} \text{sign}(\text{polar}(\tilde{\mathbf{M}}_t)) & \text{(SignMuon: after),} \\ \text{polar}(\text{sign}(\tilde{\mathbf{M}}_t)) & \text{(MuonUSign: before),} \\ \text{sign}(\text{polar}(\text{sign}(\tilde{\mathbf{M}}_t))) & \text{(MuonSign: both).} \end{cases} \quad (6)$$

The three are Algorithms A2, A4 and A5 (Appendix A.17), all transmitting one bit per parameter on the uplink. On the downlink SignMuon and MuonSign distribute a ± 1 -valued object and so cost one bit in each direction, whereas MuonUSign’s server-side $\text{polar}(\cdot)$ is dense and is transmitted at full precision.

4.1 Centralized Learning

Centrally, SignMuon is applied to the matrix-valued parameters (convolutional and fully connected weights, $n_{\text{dim}} \geq 2$), while vector parameters (biases, BatchNorm) and the final classification layer are trained with AdamW, the design of Jordan et al. (2024b), formalized in (Li and Hong 2025; Ribinin et al. 2025).

We approximate $\text{polar}(\cdot)$ by a 5th-order Newton–Schulz iteration rather than a full SVD (Algorithm A1). One further implementation choice has consequences for every experiment below.

Per-layer step sizes: the unit-gain heuristic. The methods of this paper produce step matrices from two families with *exactly* known Frobenius norms: $\|\mathbf{UV}^\top\|_F = \sqrt{r}$ for the LMO-terminated methods ($r = \min(m, n)$), and $\|\mathbf{s}\|_F = \sqrt{mn}$ for the sign-terminated ones. These scale differently with layer shape, so no single global η is appropriate for both families, or across layers within one. We fix the shape dependence a priori and tune only a shape-free base rate η_0 , setting $\eta_\ell = \eta_0 \lambda_\ell$ with

$$\lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{s}_\ell\|_F} \implies \begin{cases} \lambda_\ell^{\text{LMO}} = \sqrt{\max(1, m/n)}, \\ \lambda_\ell^{\text{SIGN}} = 1/\sqrt{\text{fan-in}}. \end{cases} \quad (7)$$

The criterion is that every layer’s update possess the same root-mean-square input–output gain, the average-case form of the spectral scaling condition (Yang, Simon, and Bernstein 2023; Large et al. 2024), so that η_0 is precisely the per-step RMS gain. We treat (7) as a heuristic and apply it uniformly for one reason above all: its LMO branch *reproduces* the aspect-ratio factor already present in the reference Muon implementation (Jordan et al. 2024b), so that a criterion accepted for one family supplies the missing counterpart for the other. Appendix A.16 gives the derivation and the measurement selecting the exponent $\frac{1}{2}$ over μP ’s 1.

4.2 Theoretical limitations

The Muon LMO direction, steepest descent in the spectral norm, does not survive sign compression, before or after the oracle: in each of the three placements the compressed step can become an *ascent* direction on a smooth objective. We read $\text{polar}(\cdot)$ off the rank-truncated SVD; at a rank-deficient argument the spectral-ball LMO is not unique, and this selection, the one our implementation makes, is part of what the counterexamples assert.

We refute the descent property on *linear* objectives, the simplest smooth functions:

$$f(\mathbf{W}) = \langle \mathbf{G}, \mathbf{W} \rangle = \text{Tr}(\mathbf{G}^\top \mathbf{W}), \quad \nabla f(\mathbf{W}) \equiv \mathbf{G}. \quad (8)$$

Gradient descent and full-precision Muon drive $f \rightarrow -\infty$ here, so a rule that instead drives $f \rightarrow +\infty$ is unambiguously ascending (Remark 1, Appendix A.8, restores Assumption 1 without moving any trajectory below).

On (8) the step is the constant matrix $\mathbf{s}(\mathbf{G})$ whatever the momentum, so momentum affects neither convergence nor divergence (Proposition 1, Appendix A.2) and

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta_t \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle \quad (9)$$

for every $\mu \in [0, 1)$ and either momentum rule. One gradient with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ thus makes f increase at *every* step, at *any* step size, with $f(\mathbf{X}_t) \rightarrow +\infty$ whenever $\sum_t \eta_t = \infty$. Such a gradient exists for every placement, at small size.

Theorems 1–3 (informal). *SignMuon, MuonUSign and MuonSign all diverge. There is a 4×4 gradient on which SignMuon ascends, and a single 5×5 gradient on which MuonUSign and MuonSign both ascend: on the corresponding linear objective (8) each method increases f at every iteration, for every step size $\eta_t > 0$, every $\mu \in [0, 1)$ and either momentum rule, with $f(\mathbf{X}_t) \rightarrow +\infty$ whenever $\sum_t \eta_t = \infty$.*

The instances are explicit and the inner products $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle$ exact rationals; Appendices A.3–A.5 state and prove the theorems and delimit the two shapes from below. For MuonUSign and MuonSign the bound is exhaustive: no shape smaller than 4×5 admits an ascending instance, and 5×5 is the smallest square one. For SignMuon, 2×2 is excluded by proof and 3×3 by an unsuccessful search. The left and centre panels of Figure 1 exhibit the ascents.

4.3 Centralized Error Feedback: EF21-SignMuon

Theorems 1–3 rule out every static placement. The classical remedy for a biased one-bit compressor is *error feedback*: store the compression error and reinject it at the next

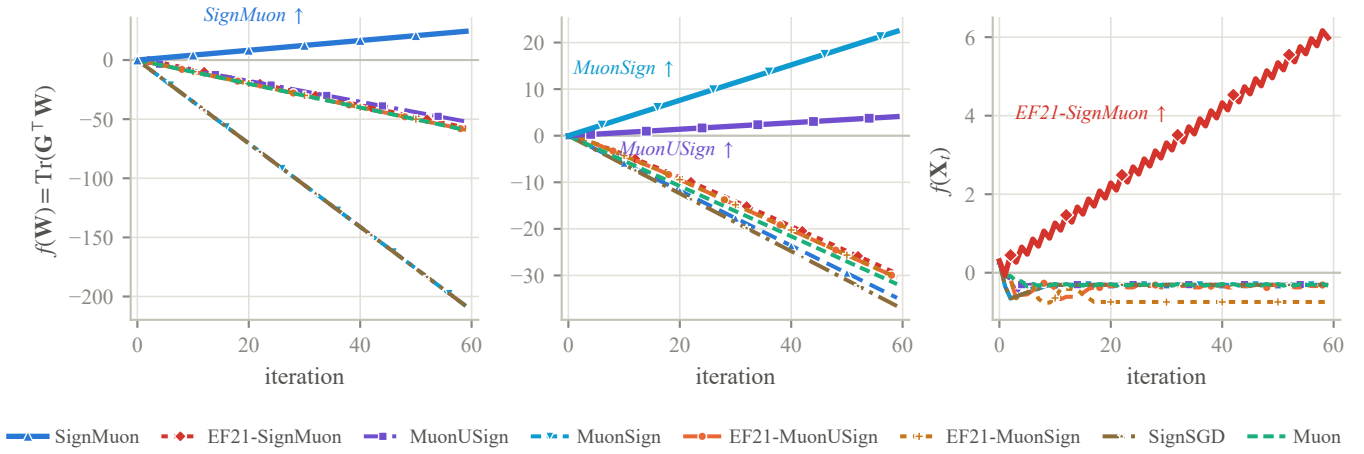


Figure 1: All eight methods on the three counterexample instances; in each panel the ascending method is drawn heavy and named. *Left*: SignMuon on the 4×4 instance of Theorem 1. *Centre*: MuonUSign and MuonSign on the shared 5×5 instance of Theorems 2–3; both panels plot the linear objective (8). *Right*: EF21-SignMuon on the 2×2 instance of Theorem 4, in normalized units with $\eta = 1$. Trajectories are momentum-free, without loss: see Proposition 1 (Appendix A.2) for the three placements and Lemma 4 (Appendix A.8) for EF21-SignMuon; the remaining baselines are shown at $\mu = 0$ only.

step, which repairs SignSGD (Karimireddy et al. 2019) and was sharpened into EF21 (Richtárik, Sokolov, and Fatkhullin 2021). Its most direct use for SignMuon keeps the geometry (the sign *after* the LMO) and error-feedbacks the oracle’s output. The resulting method, **EF21-SignMuon** (Algorithm A3), tracks an estimate $\mathbf{d}_t^{\text{est}}$ of the LMO direction $\mathbf{D}_t = \text{polar}(\tilde{\mathbf{M}}_t)$, refreshed by a scaled sign of the residual,

$$\begin{aligned} \mathbf{d}_t^{\text{est}} &= \mathbf{d}_{t-1}^{\text{est}} + \alpha_t \text{sign}(\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}), \\ \alpha_t &= \text{mean}|\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}|, \end{aligned} \quad (10)$$

and steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{d}_t^{\text{est}}$.

Error feedback on the compressed *output*, however, does not restore the descent property, and fails for *every* step size and momentum setting.

Theorem 4 (Divergence of EF21-SignMuon) *For every $L > 0$, step size $\eta > 0$, momentum coefficient $\mu \in [0, 1)$, and either momentum variant, there is an L -smooth (Assumption 2), bounded-below (Assumption 1) function $f : \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}$ on which EF21-SignMuon started at $\mathbf{X}_0 = \mathbf{0}$ diverges: for an explicit constant $c = c(f) > 0$,*

$$f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = cL\eta^2 > 0 \quad (t \geq 3), \quad (11)$$

so $f(\mathbf{X}_t) \rightarrow +\infty$. In particular, no step-size rule $\eta = \eta(L, \mu)$ using only the smoothness and momentum constants can make the method convergent.

The obstruction (proof in Appendix A.8) is specific to signing the LMO output: a *single* magnitude α_t rescales all entries’ signs at once, so a large, sign-flipping off-diagonal forces $\alpha_t = \Theta(1)$ and overwhelms the small diagonal target, whose estimate then locks onto the *wrong* sign and drives one coordinate of $\mathbf{X}_t$ to infinity. Momentum affords no remedy: Figure A1 (Appendix A.8) records the same rate for every μ and both variants.

4.4 Centralized Error Feedback: EF21-MuonUSign and EF21-MuonSign

The defect is not the shared magnitude α_t as such, since the methods below couple all coordinates through the same scalar; it is *what the estimator is asked to track*. Error feedback requires a target that settles, whereas the LMO output $\text{polar}(\tilde{\mathbf{M}}_t)$ is non-Lipschitz in the momentum and can jump by $\Theta(1)$ between rounds however small η is. Compressing the *gradient* instead leaves the estimator tracking a target that moves with the step size, which restores convergence at the same one-bit uplink cost; we adapt the EF21-Muon mechanism (Gruntkowska et al. 2025) to sign compression.

The **USign** compressor is the directional pair $(\mathcal{C}_\uparrow, \mathcal{C}_\downarrow)$ with $\mathcal{C}_\uparrow(\mathbf{Y}) = \text{mean}|\mathbf{Y}| \text{sign}(\mathbf{Y})$ the scaled sign on the uplink and $\mathcal{C}_\downarrow = I$ the identity on the downlink: one bit per parameter from client to server, a full-precision model back. **EF21-MuonUSign** pairs it with EF21 and the Muon LMO, maintaining a gradient estimator $\mathbf{g}_t^{\text{est}}$ refreshed by a contractive sign compressor applied to the residual $\Delta_t = \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\text{est}}$:

$$\alpha_t = \text{mean}(|\Delta_t|), \quad \mathbf{g}_t^{\text{est}} = \mathbf{g}_{t-1}^{\text{est}} + \alpha_t \text{sign}(\Delta_t), \quad (12)$$

and takes the descent step

$$\begin{aligned} \mathbf{X}_t &= \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t, \\ \mathbf{D}_t &= -A(\mathbf{g}_t^{\text{est}}) \approx \mathbf{U}_t \mathbf{V}_t^\top. \end{aligned} \quad (13)$$

Here $\mathbf{U}_t \mathbf{V}_t^\top$ is the polar factor of the *refreshed* estimator, not of ∇f , so no step-wise descent property is claimed, only the tracking hypothesis the guarantee requires. The sign in (12) acts on the internal compression residual alone, so the uplink remains at one bit per parameter (Algorithm A6, Appendix A.17). Figure 1 confirms that EF21-MuonUSign descends on the counterexample on which SignMuon ascends.

That hypothesis is the sole remaining requirement. Appendix A.9 shows both error-feedback methods to be *exact instances* of the EF21-Muon framework of Gruntkowska et al. (2025), the one substantive verification being that the scaled sign is a contractive compressor (Lemma 5), and so to inherit its guarantees (Theorem 5): the standard $\mathcal{O}(T^{-1/2})$ rate for $\min_{t \leq T} \mathbb{E} \|\nabla f(\mathbf{X}_t)\|_*^2$ under Assumptions 1–2, and, for EF21-MuonUSign at a plain constant step size, $\mathcal{O}(T^{-1/4})$ for $\min_t \sum_i \mathbb{E} \|\nabla_i f(\mathbf{X}_t)\|_*$ under (L^0, L^1) -smoothness (Corollaries 1–2, which also give the per-layer weighted form).

EF21-MuonSign makes both channels one-bit by applying the same scaled-sign operator to the downlink. The mechanism is a second error-feedback loop, on the *model* rather than the gradient, and it splits the method into two iterates:

$$\begin{aligned} \mathbf{X}_t &= \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t && \text{(server model),} \\ \mathbf{W}_t &= \mathbf{W}_{t-1} + \alpha_t^\downarrow \text{sign}(\mathbf{X}_t - \mathbf{W}_{t-1}) && \text{(broadcast model),} \end{aligned} \quad (14)$$

with $\mathbf{D}_t = -A(\mathbf{g}_t^{\text{est}})$ as before, and $\alpha_t^\downarrow = \text{mean} |\mathbf{X}_t - \mathbf{W}_{t-1}|$. The server model $\mathbf{X}$ takes the exact LMO step and never leaves the server; what crosses the downlink is the one-bit refresh of $\mathbf{W}$, the only model the remainder of the method observes, since gradients, momentum and the uplink residual are all computed there, whereas the guarantee bounds the gradient at $\mathbf{X}$. Under an exact downlink the two coincide and the method reduces to EF21-MuonUSign. The $\mathcal{O}(T^{-1/2})$ rate survives the second loop, though not without charge: a sign-based downlink is contractive in the Euclidean norm, whereas the spectral geometry requires contractivity in the layer norm, so the admissible step size shrinks by a further $\sqrt{r}$ in the layer rank. This is no artefact of the analysis, the scaled sign being spectrally contractive for *no* parameter at all, so that only a different compressor eliminates it (Remark 4, Appendix A.9). Section 5.3 quantifies the resulting expense; the full procedure is Algorithm A7. Both error-feedback methods incur a further charge in memory, one buffer per compressed channel: negligible on the server, potentially prohibitive on memory-limited workers.

4.5 Federated Learning

In the federated setting (2) each client computes a stochastic gradient at the broadcast model, maintains a momentum buffer, applies the Muon LMO (Algorithm A1), and uploads the elementwise sign of the result; the server takes a majority vote, $\mathbf{s}_t^{\text{agg}} = \text{sign}(\sum_j \mathbf{s}_t^{(j)})$, and sets $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\text{agg}}$. Both directions then cost one bit per parameter: $\mathbf{s}_t^{\text{agg}}$ is ± 1 -valued in every coordinate, exactly so at an odd client count, whence the server broadcasts the *vote* rather than the model, and replicas started from a common $\mathbf{X}_0$ never drift. This is a property of the vote, not of the compressor (Bernstein et al. 2019).

The error-feedback variants relocate the LMO from the clients to the server: each client sign-compresses the residual between its momentum and its gradient estimator and sends $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$; the server aggregates, refreshes a global estimator, and applies the LMO. The additional per-layer scalar leaves the uplink at ≈ 1 bit per parameter, but the server-side

quantity is now dense, being a scaled average of signs followed by the real-valued $\text{polar}(\cdot)$ of that average, so the vote argument lapses and the full model must be broadcast unless a second error-feedback loop compresses it, which is the addition EF21-MuonSign makes. The complete procedures are Algorithms A8–A9.

5 Experiments

What a controlled problem shows first. A deterministic L -smooth convex quadratic (Appendix A.11) permits us to measure the scalar upon which the descent lemma turns: the alignment $\rho_t = \langle \nabla f, \mathbf{D}_t \rangle / (\|\nabla f\|_F \|\mathbf{D}_t\|_F)$, which Theorems 1–3 drive negative by construction. On random instances it never becomes negative: all three placements keep ρ_t well clear of zero, so the counterexamples are not a configuration ordinary data reproduces. That is direct evidence for reading them as worst-case rather than predictive, and it accounts for the network results below running contrary to them. The one method whose ρ_t does dip negative is EF21-MuonSign, the one carrying a proof: error feedback converges through its estimator, not through per-step descent. Two readings carry forward. SignMuon and SignSGD emit steps of *identical* Frobenius length, so whatever separates them is alignment, not step length. And every method here but SGD takes a norm-fixed step, which at a constant rate settles into a ball of radius $\eta \|\mathbf{s}\|_F$ rather than converging, so a fixed-target iteration count ranks accuracy *floors*, not rates.

5.1 Centralized Learning

We compare all six matrix-aware 1-bit methods against Muon, SignSGD, SGD and Adam on CIFAR-10 (Krizhevsky, Hinton et al. 2009), using a ResNet-18 adapted to low-resolution images, for 75 epochs with η_0 cosine-annealed to zero. The only tuned hyperparameter is η_0 , selected per method on a held-out split and interpreted identically across methods by the unit-gain rule (7). Centralized numbers are means over three seeds and federated numbers over five; differences smaller than the seed spread are not claimed as results. Grid, endpoint rule and schedules are in Appendix A.10.

The top three rows of Table 1 lie within one seed spread, so we claim only that sign-after *matches* Muon at one bit per parameter, error feedback exacting nothing measurable in addition. What does resolve is the comparison against SignSGD, which expends the same budget without the LMO: the margin measures the value of the matrix geometry, not of the compressor, as the alignment reading predicts.

Within the family the ordering exceeds the noise, after before both sides, and the training loss orders the methods identically (Figure A5), so the separation is one of optimization rather than of regularization. The threshold column distinguishes them far more sharply: SignMuon attains 90% in 8.3 epochs against 18.8–21.7 for SGD, SignSGD and Adam, a factor of 2.3–2.6, and orthogonalization’s $\sim 18\%$ per-epoch overhead still leaves a 1.9 – $2.2\times$ wall-clock margin.

5.2 Federated Learning

Here the bottleneck is transmitting full 32-bit gradient matrices. We compare the same ten methods on CNN2 (two con-

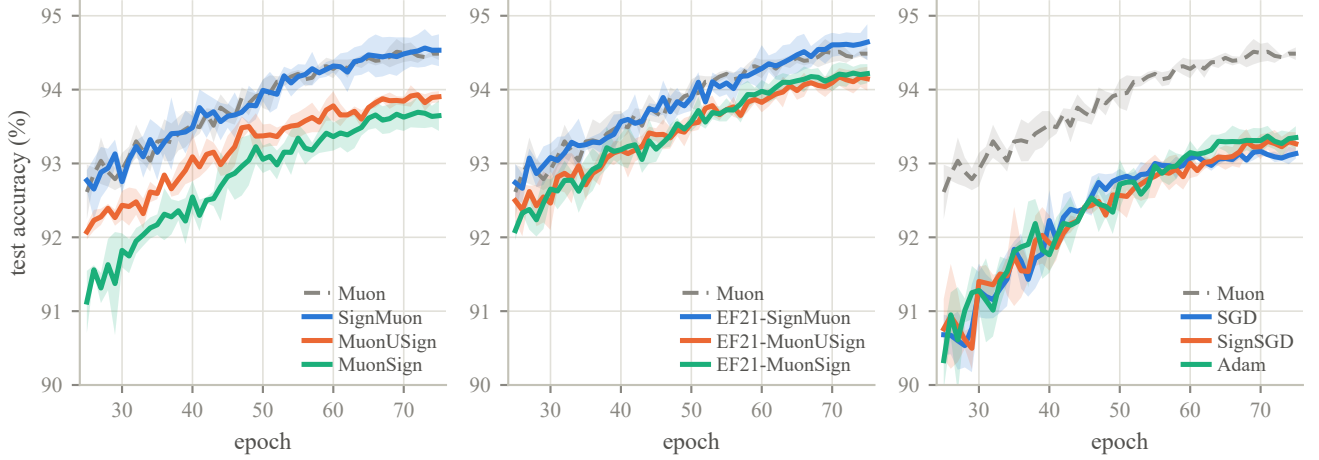


Figure 2: Centralized ResNet-18 on CIFAR-10: test accuracy from epoch 25, at each method’s selected η_0 (Table 1). Panels group the three sign placements (left), the EF21 variants (centre) and the uncompressed baselines (right); Muon is the gray dashed reference throughout, bands ± 1 standard deviation over three seeds.

Method	η_0	Test acc (%)	Ep. to 90%
EF21-SignMuon	0.01	94.64 ± 0.24	8.7
SignMuon	0.02	94.53 ± 0.22	8.3
Muon	0.05	94.49 ± 0.09	7.7
EF21-MuonSign	0.01	94.22 ± 0.13	10.3
EF21-MuonUSign	0.2	94.15 ± 0.16	8.7
MuonUSign	0.02	93.90 ± 0.08	10.0
MuonSign	0.005	93.65 ± 0.21	15.0
Adam	0.0005	93.35 ± 0.15	18.8
SignSGD	0.002	93.27 ± 0.16	19.0
SGD	0.02	93.13 ± 0.07	21.7

Table 1: Centralized ResNet-18 / CIFAR-10, 75 epochs. Test accuracy is the mean $\pm$ standard deviation over three seeds of the last five epochs; “Ep. to 90%” is the mean epoch at which it first reaches 90%. Rules separate the statistically tied methods.

volitional blocks with BatchNorm and an MLP classifier), with CIFAR-10 partitioned homogeneously across $N = 11$ clients and no local parameter steps. That is the scale at which the sign-based methods separate, and it is odd by design, since client messages are ± 1 valued and an odd N leaves the majority vote decisive in every coordinate. Local models are discarded each round, so BatchNorm remains at its initialization statistics; rates are tuned per method on a split held out *before* the client partition, so that no placement is handicapped by its step size and no test image is scored during selection (Appendix A.10). Two schemes cover all six, worker-side LMO (Algorithm A8) and server-side LMO (Algorithm A9); Table A9 records which compressor pair determines each.

Federation separates the methods far more, and Table 2 admits three readings. First, SignMuon attains Muon’s quality at one bit per parameter in *both* directions, within the

Optimizer	η_0	Up	Down	Rds. to 80%	Test acc (%)
Muon	0.1	32	32	260	85.99 ± 0.26
SignMuon	0.05	1.09	1.09	300	85.81 ± 0.25
EF21-SignMuon	0.05	1.09	32	600	85.16 ± 0.15
MuonUSign	0.05	1.09	32	540	84.62 ± 0.17
EF21-MuonSign	0.05	1.09	1.09	980	83.99 ± 0.28
EF21-MuonUSign	0.05	1.09	32	1000	83.75 ± 0.21
SignSGD	0.02	1.09	1.09	1080	81.83 ± 0.43
MuonSign	0.05	1.09	1.09	1300 [†]	80.68 ± 1.88
SGD	0.1	32	32	1367 [†]	79.79 ± 1.55
Adam	0.001	32	32	—	77.12 ± 1.04

Table 2: CIFAR-10 federated learning on CNN2, ordered by accuracy: $N = 11$ clients, homogeneous split, 2000 rounds at batch 192 per client, momentum 0.9. Test accuracy is the mean of the final five evaluations over five seeds, $\pm$ one standard deviation *across* seeds. Up and Down are mean bits per model parameter per round in each direction (Appendix A.13). [†] mean over the seeds reaching 80% within the budget (4/5 for MuonSign, 3/5 for SGD); Adam reached it on none.

seed spread; at that fixed round trip, signing the LMO direction rather than the gradient is worth several accuracy points over SignSGD, considerably more than centrally. Second, five seeds resolve the three placements, which single-seed evidence had shown as coincident: they order after, before, both sides, the last below SignSGD.

Third, and least comfortably, the placement that performs best is the one the theory condemns: sign-after is the placement Theorem 1 makes diverge, and error feedback repairs it for no (L, η, μ) whatever (Theorem 4), yet the sign-after methods are the strongest compressed ones here, centrally, and on nanoGPT (Section 5.3). The two variants carrying an

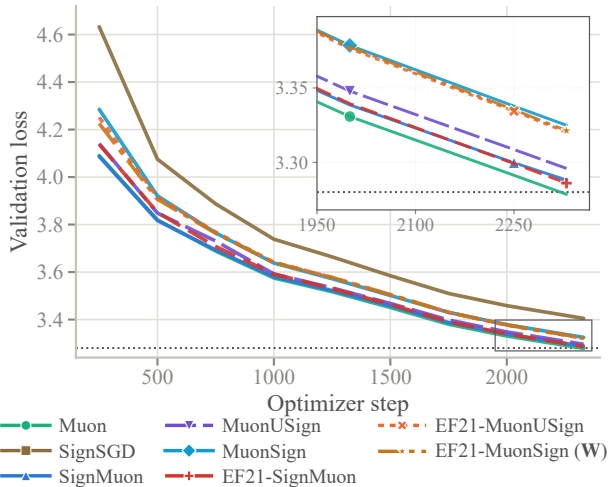


Figure 3: NanoGPT speedrun, 8xH100: validation loss against optimizer step, single seed; dotted is the speedrun target 3.28. EF21-MuonSign is drawn at its broadcast model $\mathbf{W}$. The inset magnifies the boxed tail, where SignSGD is above the range shown; exact values are in Table 3.

Method	Step	η_0	Val. loss	Steps to 3.35
Muon (record #40)	LMO	0.06	3.2785	1.00×
EF21-SignMuon	LMO	0.06	3.2860	1.02×
SignMuon	SIGN	0.03	3.2881	1.02×
MuonUSign	LMO	0.06	3.2959	1.05×
EF21-MuonUSign	LMO	0.06	3.3203	1.13×
EF21-MuonSign ($\mathbf{W}$)	LMO	0.06	3.3213	1.14×
MuonSign	SIGN	0.03	3.3249	1.14×
SignSGD	SIGN	0.03	3.4049	—
EF21-MuonSign ($\mathbf{X}$)	LMO	0.06	5.5198	—

Table 3: NanoGPT after 2330 steps (611M tokens), single seed. Record #40 reports 3.2780 ± 0.0009 over five seeds, so differences below ~ 0.003 are noise. “Steps to 3.35” is relative to Muon. The last row is EF21-MuonSign’s server model $\mathbf{X}$, the same run as the broadcast model $\mathbf{W}$ above.

unconditional $\mathcal{O}(T^{-1/2})$ guarantee trail SignMuon by several seed spreads, so the ordering is not noise. The guarantee exacts accuracy at this scale, and the alignment measurement above indicates why it may.

The second compression is not the source of that expense: EF21-MuonSign stands marginally *ahead* of the uplink-only EF21-MuonUSign, so at CNN2’s scale the downlink loop is costless and the placement alone separates the EF21 family from SignMuon. All methods are scored at the server model $\mathbf{X}_t$, which for EF21-MuonSign is not the model the clients hold (14); that the two nonetheless track each other here, and not at transformer width, is the subject of Section 5.3 and Remark 3 (Appendix A.9).

5.3 Language Modelling

We test where the matrices are large enough for the layer-rank dependence of Corollary 1 (Appendix A.9) to take effect: the modded-nanoGPT speedrun (record #40), a 12-layer transformer with 768×3072 hidden matrices, trained on 611M FineWeb tokens on 8xH100 (Jordan et al. 2024a). Each method replaces the record’s Muon on the hidden matrices and gates, leaving all else untouched; our Muon arm is record #40’s own optimizer and reproduces its published result within seed noise, validating the port. Rates are fixed a priori by the unit-gain rule at one η_0 per *family* rather than tuned per method: every contrast is matched-hyperparameter, at the cost of not reporting each method’s best loss. Runs are single-seed; Table 3 states the reference’s seed noise (Appendix A.15).

Three conclusions follow (Table 3, Figure 3). First, the matrix-aware 1-bit methods transfer to language modelling: every one of them improves on SignSGD by at least 0.08 in validation loss, some eighty times the reference’s seed standard deviation, while the best trail full-precision Muon by 0.008, at equal wall-clock. Second, error feedback is inexpensive but not costless: the transition MuonUSign $\rightarrow$ EF21-MuonUSign requires $\sim 8\%$ more steps, the price of steering the LMO with an estimator that lags its target. EF21-SignMuon, whose target polar($\bar{\mathbf{M}}_t$) is orthogonal and hence isotropic, forfeits almost nothing and is the strongest compressed method here, the placement the theory condemns again finishing ahead.

Third, the bidirectional method’s two models (14) separate at this width. Its broadcast model $\mathbf{W}$, at which every gradient is evaluated, is indistinguishable from EF21-MuonUSign, so the compressed downlink is costless where training occurs; the server model $\mathbf{X}$, the iterate the guarantee bounds, instead settles ≈ 2.2 nats above it, a persistent offset rather than a divergence, which Appendix A.15 localizes to the single zero-initialized $\mathbf{c}_{\text{proj}}$ layer. This is the mechanism of Remark 3 rather than a tuning failure: only a spectrally contractive downlink compressor eliminates it (Remark 4).

6 Conclusion

A sign step and a spectral step are each an LMO, each sound alone; composed, they constitute an LMO for no norm, and the descent property is forfeited in the composition rather than in either factor. On explicit instances, minimal in shape, *every* placement of the sign around the oracle converts the update into an ascent direction on a linear objective, at any step size and any momentum; and error feedback on the oracle’s *output* repairs none of them, its single shared magnitude being unable to track a target that may jump by a constant.

Compressing before the oracle rather than after it does succeed: error-feedbacking the gradient leaves the estimator pursuing a target that moves at the step size, and the resulting methods are exact instances of an existing framework, inheriting its $\mathcal{O}(T^{-1/2})$ rate at a one-bit uplink and, with a second loop on the model, in both directions. The second bit is nearly costless in practice but not in theory: a sign compressor contracts in the Euclidean geometry and in no layer norm, so the step size forfeits a factor in the layer rank,

manifest as the separation of the two models at transformer width.

Theory and experiment order the methods oppositely: on every task we measure, the best compressed method is the placement we prove unrepairable, while the provably convergent variants trail it. A controlled problem indicates why: the alignment our constructions drive negative remains comfortably positive on random instances, so the guarantee insures against an event those problems never generate. What we preclude is a guarantee holding *everywhere*, not one holding under a realistic condition, and the identification of that condition is the open question. Two narrower ones remain: the (L^0, L^1) rate under a compressed downlink, and whether any one-bit compressor is layer-norm contractive.

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